

HONGJING ZHU

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RESEARCH INTERESTS

An undergraduate student working on robot learning and embodied intelligence, with a focus on representation learning, knowledge and experience reuse, and autonomous decision-making and adaptation. I am interested in how robots can learn transferable representations and policies that autonomously leverage feedback from physical interactions. My goal is to enable robots to draw on prior experience, continually refine their understanding of tasks and environments, and adjust their plans and actions to reliably complete long-horizon tasks and adapt to new situations.

EDUCATION

Shenzhen Technology University

Sep 2024 – Present

B.Eng. Candidate, Unmanned Aircraft Systems Engineering | GPA: 3.0/4.0

INTERNSHIP EXPERIENCE

BrainCo — Dexterous Hand Algorithm Intern

Feb 2026 – Jun 2026

Open-source contribution: [RevoLab](#)

HORA/RMA In-Hand Rotation — Task Modeling, Training, and Deployment

- ★ Adapted HORA/RMA to ball and cylinder rotation on the Revo3 dexterous hand, building Isaac Lab environments and a two-stage pipeline for teacher training with privileged information and history-based adaptation.
- ★ Designed rotation and grasp-stability rewards with domain randomization to account for variations in object dynamics and actuator properties.
- ★ Built an ONNX-based pipeline for policy export and closed-loop hardware deployment; demonstrated the trained policy at BrainCo's ICRA 2026 exhibition booth.

Dynamic Handover — Dual-Arm Task Modeling and Policy Training

- ★ Adapted dynamic throwing and catching to the RevoTron dual-arm platform; formulated a command-conditioned task for training a single policy across handover directions.
- ★ Implemented reward shaping for throwing, receiver tracking, and stable catching; built parallel training and evaluation workflows and refined task configurations through failure analysis.

Robot Policy Training — Model Adaptation and Evaluation

- ★ Built a shared data-to-policy pipeline using LeRobot for multimodal data standardization, connecting preprocessing, policy training, and offline evaluation.
- ★ Adapted ACT, $\pi 0.5$, and GR00T N1.6 training and evaluation workflows, supporting LoRA fine-tuning and multi-GPU training while aligning datasets with model-specific observation and action requirements.

PUBLISH

Human-Aware Physical Intelligence for Adaptive Human–Robot Collaboration: Mechanisms, Applications and Challenges

2026.09

Hongjing Zhu, Haotong Luan*, et al. *Co-first authors. [Submitted to Smartbot](#).*

HONORS & AWARDS

2025 AI Robot Simulation Development Challenge — Undergraduate First Prize

Jan 2026